

WASP-36b

R-1250

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210206 · created 2026-07-31T11:46:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459251.8218 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.048
Transit depth (Rp/Rs)^2	1.78 +/- 1.29 [%]
Orbital Inclination (inc)	87.94 +/- 2.93
Airmass coefficient 1 (a1)	60.67 +/- 39.0
Airmass coefficient 2 (a2)	-2.88 +/- 1.54
Scatter in the residuals of the lightcurve fit is	78.41 %
Best Comparison Star	#1 - [613, 58]
Optimal Aperture	1.11
Optimal Annulus	7.5
Transit Duration (day)	0.0898 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.048 vs 0.1368 (deviation 0.08σ)
Duration fit vs published	0.0898 d vs 0.0758 d (deviation 1.5σ)
Mid-time O–C	-9.7 min
Depth SNR	2.8
Residual scatter	78.41 %

Light curve

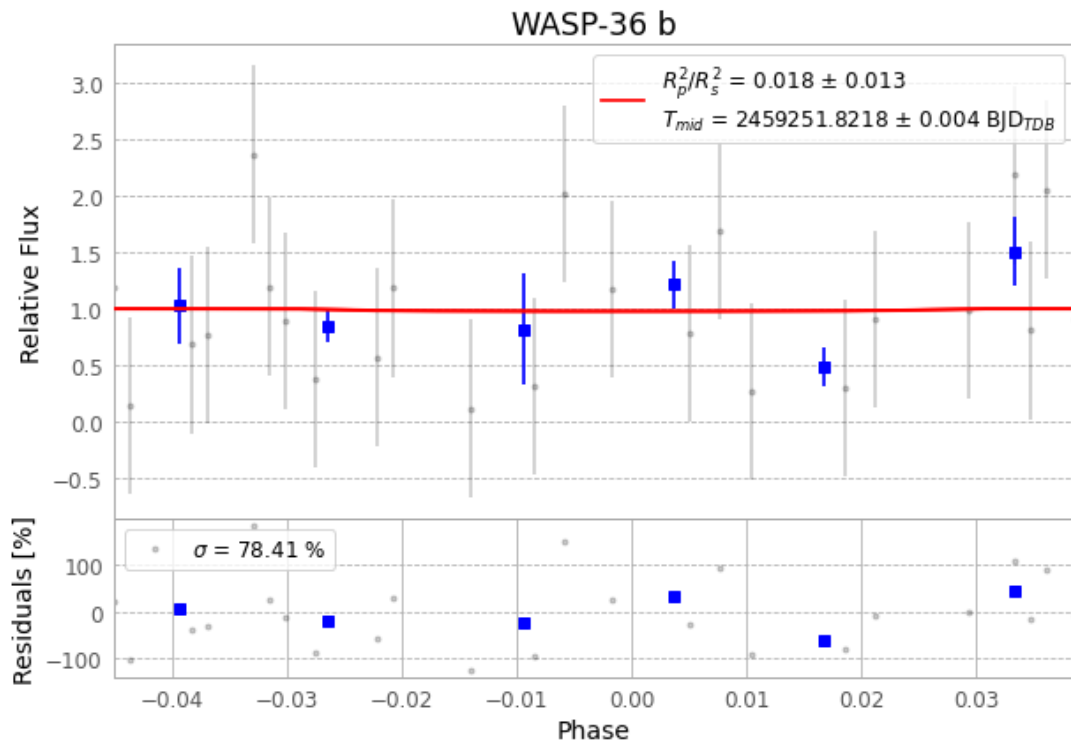


Figure 1 — FinalLightCurve_WASP-36 b_2021-02-05.png (SHA-256 09bd11208947eb80...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [509, 445], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5907a6f23697205b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), WASP-36210206055412.FITS → WASP-36210206085941.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210206.log (443fc920be60...), FinalLightCurve_WASP-36 b_2021-02-05.png (09bd11208947...), FinalLightCurve_WASP-36 b_2021-02-05.csv (ed577e256d43...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:46Z.